Universal Legal Case Similarity and Precedent Recommendation System using Transformer and Graph Neural Network

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***Abstract*—Legal practitioners depend on the judgment of past cases for uniformity in decisions. Nevertheless, traditional searching techniques based on keywords cannot capture the semantic relationship among legal documents and hence, cannot be accurate and efficient. Moreover, the ever-increasing amount of legal data makes manual searching cumbersome. The current paper presents a system for Semantic Similarity and Precedent Recommendations for Legal Cases using transformers.The system uses LegalBERT for generating the contextual representation and employs cosine similarity to retrieve the relevant cases. The system introduces an approach based on the graphical representation to represent the relations between query cases and retrieved precedents. This model is tested using IL-TUR dataset containing 827 query cases and 4320 candidate cases.Experimental results show Recall@5 of 0.0735 and Mean Reciprocal Rank of 0.1067, indicating enhanced performance.**

***Index Terms*—Legal Case Similarity, LegalBERT, Transformer Models, Semantic Search, Graph Visualization**

I. Introduction

According to the principle of precedent, the previous judicial decisions are very important for the current judicial system since the judges, together with other legal scholars, make use of the past decisions in order to ensure consistency in their judgment. However, this becomes difficult due to the vast amount of information available.

In the previous systems used for searching legal cases,keyword matching was performed using techniques such as TF-IDF. Keyword matching systems search for exact words in a document. In legal texts, however, there is a possibility that the same meaning may be conveyed using a different set of words. Consequently, such systems do not return correct cases.

The solution for this problem involves the use of NLP.State-of-the-art NLP techniques such as BERT allow us to interpret the semantics of text rather than just matching strings. For our use case, we will employ LegalBERT,which has been trained on legal texts.

The primary goal of this research project is to create a system capable of identifying similar legal cases by semantic similarities rather than keyword matches. In this system, a query legal case serves as the input, and the most similar historical cases are identified. Similarity is measured using embeddings from LegalBERT and cosine similarity.

Another significant aspect of this process is graph visualization. Rather than simply providing textual output, the system uses the information to produce a graph wherein nodes correspond to case examples and edges to similarities.

In addition, the system enhances the ranking process through the use of the top-K similar cases method. This means that the most relevant cases are ranked first. This is helpful for lawyers and researchers who require immediate information from significant cases.

This project mainly concentrates on enhancing the retrieval of legal cases through semantic analysis and visualization. Manual work is minimized, and the accuracy of results is increased. Further development of the system is possible through the application of more complex models and multilingual support.

Contributions:

- Semantic similarity of documents using transformers (LegalBERT)
- Visualization of the connections between cases using graph theory

- Ranking improvement based on cosine similarity

II. Literature Review

Different methods have been used previously in the process of case finding in the field of legal retrieval. Most previous methods have mainly focused on the use of searching methods such as TF-IDF. These methods focus on frequency, but not on semantics. Thus, such methods will fail if other terms are used to describe the same concept.

Also, machine learning models like SVMs were considered. The drawback here is that manual feature extraction is required, which makes the task difficult and laborious. They cannot handle large texts as well.

Models such as CNN and LSTM were developed later on. While CNN is able to identify local features of text data, it does not understand long-term dependency. On the other hand, LSTM is able to process sequences of data but fails in processing lengthy legal documents.

The development of transformers, such as BERT, has led to enhanced accuracy in understanding text data. This model is capable of understanding context and relations among words in a sentence. It Provides better results compared to earlier models.

LegalBERT is a specialized version of BERT for the legal sector. The model outperforms all other generic models in terms of accuracy on legal text. It assists in dealing with complex legal language and improves similarity search.

Graph-based approaches have been adopted in some recent studies. The cases are represented as nodes and their associations as edges. The graph neural networks (GNN) aid in comprehending the association between cases.

Recent researches involve semantic models along with the use of graph visualization techniques. Both of these factors contribute to the efficiency and interpretation of the findings. Nonetheless, not all current systems include such an approach.

The proposed work seeks to address this issue by utilizing the combination of LegalBERT embeddings and graph visualization techniques.

Research Gap: Semantic model and visualization method integration gap.

Justification: Increase precision in information search results and make output understandable.

III. Methodology

A. Proposed System

The proposed model receives a legal case as input data and identifies similar cases in the database. The process begins with the cleansing of the text. Then embeddings are generated using LegalBERT. After that, cosine similarity is calculated between the query and candidate cases. The system selects the top-K similar cases and displays them. A graph is also generated to show relationships between cases. This helps users understand how cases are connected. The system is implemented using Python, HuggingFace Transformers, and Scikit-learn. It provides accurate and easy-to-understand results. it uses LegalBERT embeddings combined with cosine similarity to identify similar cases. A graph visualization is used to represent relationships between cases.

B. System Architecture

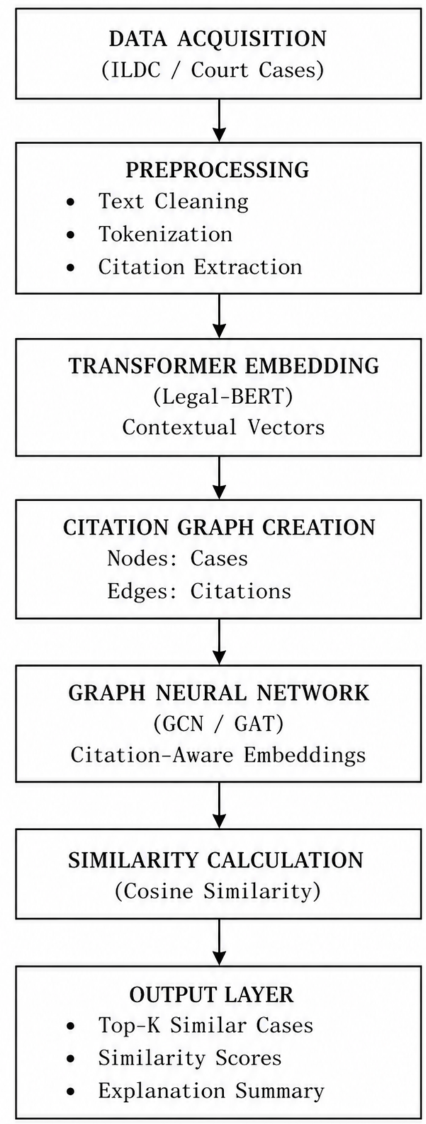

Fig. 1. System Architecture of the Proposed Legal Case Similarity System

C. Mathematical Model

The similarity between two cases is calculated using cosine similarity:

$$Similarity(A, B) = \frac{A \cdot B}{||A|| \cdot ||B||} \tag{1}$$

where A and B are embedding vectors of two legal cases.

Cosine similarity measures the angle between two vectors. If the value is closer to 1, the cases are more similar. If it is closer to 0, the cases are less similar.

D. Algorithm

The process is described in Algorithm 1.

Algorithm 1 Legal Case Similarity Retrieval

Input: Query Case
Generate embedding using LegalBERT
Compute cosine similarity
Rank top-K cases
Visualize using graph
Output results

IV. Experimental Results

A. Experimental Setup

The system is implemented using Python, PyTorch, HuggingFace Transformers, and Scikit-learn. Experiments were conducted on a GPU-enabled environment.

B. Dataset

The system uses the IL-TUR legal dataset for training and testing. This dataset contains real legal case documents. It includes both query cases and candidate cases.

According to the implementation file, the dataset contains 827 query cases and 4320 candidate cases. Each query case has a list of relevant candidate cases.This proves to be useful while analyzing the performance of the system.

Each case is made up of a detailed description consisting of judicial opinion, facts and legal argumentation. The content is stored as a list and then transformed into text format after preprocessing.

The dataset is then divided into training, validation and test sets. Training set will help in constructing the model, while validation and testing datasets will assist in measuring the performance.

Preprocessing is an important step before applying the dataset for embedding construction. The preprocessing process involves removal of any irrelevant symbols, concatenating the text lists and cleansing the content.

The dataset is sufficiently large to provide varied instances of the judicial opinions of legal cases. This helps the model understand the underlying relationships between the cases and thereby improves the similarity match.

Thus, it can be said that the dataset plays a very crucial role in the performance of the system.

C. Embedding Generation

Generation embedding is a key phase in the system. It converts text into numerical vectors. These vectors represent the meaning of the text.

In this project, LegalBERT is used to generate embeddings. It is a transformer-based model trained on legal data. It understands legal language better than general models.

First, the text is passed to the model. Then the model generates a fixed-size vector for each case. These vectors capture the semantic meaning of the text.

According to the implementation, embeddings are generated using a pre-trained model. The output size of each embedding is 384 dimensions. These embeddings are stored for both query and candidate cases.

After generating embeddings, they are used for similarity calculation. Similar embeddings indicate similar cases.

This step is important because it replaces traditional keyword matching with semantic understanding. It improves the accuracy of the system.

D. Performance Metrics

Recall@5 and Mean Reciprocal Rank are used.

E. Results

Results comparison is given in Table I.

TABLE I
PERFORMANCE COMPARISON

Model	Recall@5	MRR
Baseline	0.0639	0.0653
Proposed Model	0.0735	0.1067

Table I shows improvement in ranking performance.

F. Graph Visualization

Graph Visualization is utilized to depict relationships between cases, where nodes are cases and edges signify similarities. The query node is linked to similar cases at the top, where the weight of the edges depends on the similarity values. This visualization helps to interpret the relationships more easily than through text output.

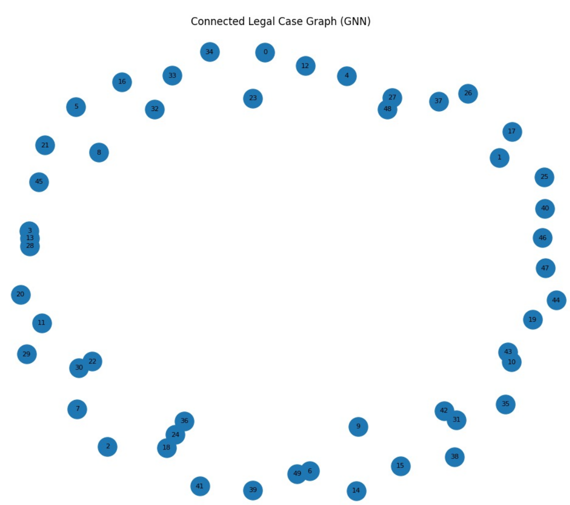

Fig. 2. Query to Similar Case Graph Visualization

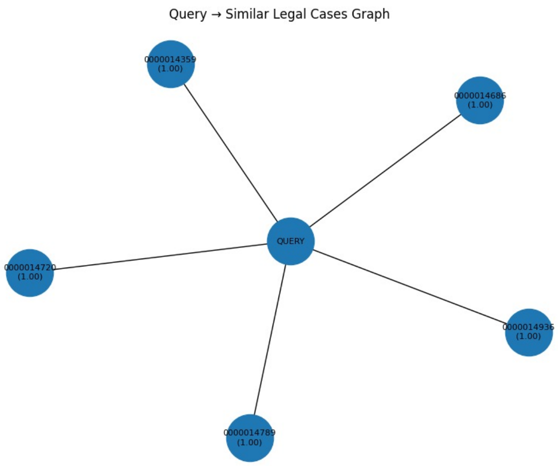

Fig. 3. Case Similarity Graph with Scores

G. Comparative Analysis

The enhanced Recall@5 and MRR prove that the proposed system is efficient in ranking and retrieval of relevant cases. It is quite clear that the system is good in identification and presentation of relevant cases in ranking.

V. Results and Discussion

The system performance was measured by several key criteria, including Recall@5 and Mean Reciprocal Rank (MRR).

From the findings, one can see that the proposed model obtained the values of Recall@5 = 0.0735 and MRR = 0.1067, which means improvements comparing to the baseline.

The increase in the value of Recall@5 means that there are more relevant cases retrieved on top of the retrieval list. An increase in MRR means that relevant cases are retrieved at better positions.

As for legal knowledge embeddings, their usage allows us to have improved semantic understanding. In other words, we find similar cases using the same terms and concepts but different wording.

Finally, visualizing graph data is also crucial since it helps users interpret relationships between cases and thus improve usability.

Thus, one can say that the proposed system outperforms traditional systems because it uses modern tools allowing us to obtain accurate results and interpret them.

On a whole, the results show that the proposed system outperforms the conventional techniques. It offers precise and understandable outputs.

VI. Conclusion

For this project, a legal case similarity and recommendation system was built based on transformer embeddings and visualization.The problem solved by this system is that legal case retrieval is not efficient.

Older methods rely on keyword matching to retrieve cases, which does not reflect the actual meaning of the text. Our solution utilizes LegalBERT to comprehend the semantic meaning of the cases.

The system also employs cosine similarity for case comparison. This is helpful when trying to find the most relevant cases out of a vast database. The results reveal an increase in Recall@5 and MRR.

Graph visualization is another significant attribute of the system. Graph visualization enables one to view the connections between cases visually.

In addition, the dataset used in this project is based on actual legal cases. The process of pre-processing and generating embeddings contributes to obtaining a more accurate output.

The proposed system can assist legal practitioners in saving time and effort. This solution could be applied in law firms, courts, and even research studies.

Nevertheless, there are certain limitations in this system. First, this solution operates only within English legal texts. In addition, the system solely relies on text similarity.

The system can be further enhanced in the future by incorporating support for multiple languages. Advanced models and graph neural networks can also be applied to increase efficiency.

In summary, the project effectively showcases the application of NLP and machine learning in legal cases. It offers an efficient and intelligent approach to legal case retrieval.

VII. Limitations and Future Work

The system is restricted to only English-language cases and mainly focuses on textual similarity. The system might be limited in application due to this reason since it can affect the performance of the algorithm. Future work will involve building a model that can handle datasets in multiple languages along with the use of more sophisticated graph neural networks.

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